

2.10.37

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Question

If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three non-coplanar vectors, then find the value of

$$(\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot [(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{c})].$$

Solution

We know that the scalar triple product can be written as a determinant:

$$[\mathbf{p} \ \mathbf{q} \ \mathbf{r}] = \mathbf{p} \cdot (\mathbf{q} \times \mathbf{r}) = \begin{vmatrix} \mathbf{p} & \mathbf{q} & \mathbf{r} \end{vmatrix}.$$

$$(\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot [(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{c})] = \begin{vmatrix} \mathbf{a} + \mathbf{b} + \mathbf{c} & \mathbf{a} + \mathbf{b} & \mathbf{a} + \mathbf{c} \end{vmatrix}. \quad (1)$$

$$\begin{vmatrix} \mathbf{a} + \mathbf{b} + \mathbf{c} & \mathbf{a} + \mathbf{b} & \mathbf{a} + \mathbf{c} \end{vmatrix} = \begin{vmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix}. \quad (2)$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = -1. \quad (3)$$

Therefore

$$(\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot [(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{c})] = \begin{vmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \end{vmatrix} \cdot (-1) \quad (4)$$

$$= -[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]. \quad (5)$$

$$\boxed{(\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot [(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{c})] = -[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]}$$